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# Chương 3: Biểu diễn Fourier của TH và HT.

## Bài tập chương 3.1.

**Bài 3.1.** Cho $x(t) = 3\cos\left(\frac{\pi t}{2} + \frac{\pi}{4}\right)$.

1. Xđ biểu diễn Fourier của tín hiệu
2. Vẽ phổ biên độ, phổ pha của $x(t)$.

Giải:

$w_0 = \frac{\pi}{2}$ $\quad T = 4$.

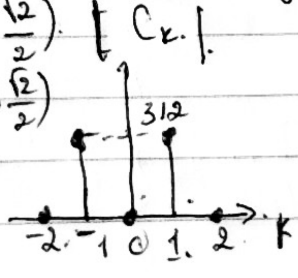
$$\Rightarrow x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\frac{\pi}{2}t} \quad \text{<→ Biến đổi Fourier>}$$

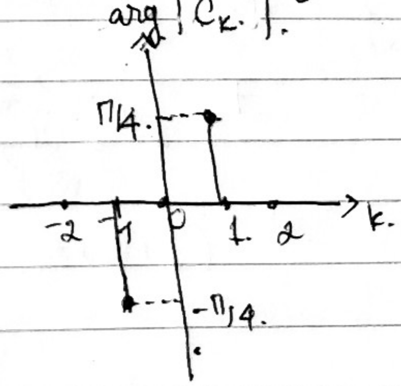
Sd CT Euler $\Rightarrow x(t) = 3 \cdot \dfrac{e^{j\left(\frac{\pi}{2}t + \frac{\pi}{4}\right)} + e^{-j\left(\frac{\pi}{2}t + \frac{\pi}{4}\right)}}{2}$

$$= \frac{3}{2} e^{j\frac{\pi}{4}} \cdot e^{j\frac{\pi}{2}t} + \frac{3}{2} e^{-j\frac{\pi}{4}} \cdot e^{-j\frac{\pi}{2}t}.$$

$$C_k = \begin{cases} \frac{3}{2} e^{-j\frac{\pi}{4}} & k = -1 \\ \frac{3}{2} e^{j\frac{\pi}{4}} & k = 1 \\ 0 & k \text{ khác} \end{cases} \Rightarrow C_k = \begin{cases} \frac{3}{2}\left(\cos\frac{-\pi}{4} + j\sin\frac{-\pi}{4}\right) & k = -1 \\ \frac{3}{2}\left(\cos\frac{\pi}{4} + j\sin\frac{\pi}{4}\right) & k = 1 \\ 0 & k \text{ khác} \end{cases}$$

$$\Rightarrow C_k = \begin{cases} \frac{3}{2}\left(\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}\right) \\ \frac{3}{2}\left(\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right) \\ 0 \end{cases}$$




$$\Rightarrow |C_k| = \begin{cases} \frac{3}{2} & k = \pm 1 \\ 0 & \text{khác} \end{cases}$$

**Bài 3.2:** Tìm biểu diễn Fourier của tín hiệu

$$x(t) = 2\sin(2\pi t - 3) + \sin(6\pi t).$$

Giải: $\quad T_1 = 1$; $\; T_2 = \frac{1}{3}$.

$T = BCNN(T_1, T_2) \Rightarrow T = 1. \Rightarrow x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk2\pi t}$.

$\Rightarrow w_0 = 2\pi$.

sd CT Euler:

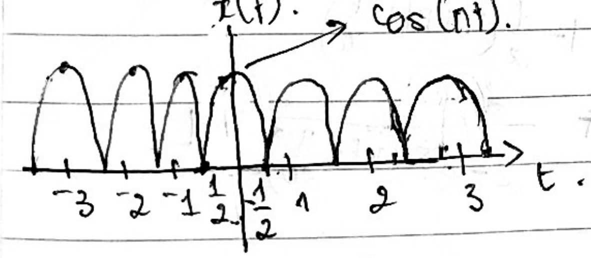
$$\Rightarrow x(t) = 2\cdot\frac{1}{2j}\left(e^{j(2\pi t-3)} - e^{-j(2\pi t-3)}\right) + \frac{1}{2j}\left(e^{j6\pi t} - e^{-j6\pi t}\right).$$

$$= \frac{1}{j} e^{2\pi t j} e^{-3j} - \frac{1}{j} e^{-2\pi t j} e^{3j} + \frac{1}{2j} e^{j6\pi t} - \frac{1}{2j} e^{-j6\pi t}$$

$$= \frac{e^{-3j}}{j} e^{2\pi t} - \frac{e^{3j}}{j}\cdot e^{2\pi t(-1)} + \frac{1}{2j} e^{j2\pi t\cdot 3} - \frac{1}{2j} e^{j2\pi t\cdot(-3)}.$$

$$\Rightarrow C_k = \begin{cases} \dfrac{e^{-3j}}{j} & k=1. \\ -\dfrac{e^{3j}}{j} & k=-1. \\ \dfrac{1}{2j} & k=3. \\ -\dfrac{1}{2j} & k=-3. \\ 0 & k \text{ khác} \end{cases}$$

Bài 3.8 Tìm các hệ số FS của dạng sóng:




$\Rightarrow x(t) = |\cos(\pi t)|.$

$\Rightarrow T=1 \quad w_0 = 2\pi.$

$$\Rightarrow C_k = \frac{1}{2}\int_{-1/2}^{1/2} \left(e^{jwt} + e^{-jwt}\right) e^{-jkw_0 t}\, dt.$$

$$= \frac{1}{2}\int_{-1/2}^{1/2} \left(e^{j\pi t} + e^{-j\pi t}\right) e^{-jk2\pi t}\, dt.$$

$$= \frac{1}{2}\cdot\int_0^1 \left[e^{j\pi(1-2k)t}\, dt + e^{-j\pi(1+2k)t}\, dt\right].$$

$$= \frac{1}{2}\cdot\left[\frac{1}{j\pi(1-2k)}\cdot e^{j\pi(1-2k)t}\Big|_{-1/2}^{1/2} + \frac{-1}{j\pi(1+2k)} e^{-j\pi(1+2k)t}\Big|_{-1/2}^{1/2}\right.$$

$$= \frac{1}{2}\left( \frac{e^{\frac{j\pi(1-2k)}{2}} - e^{\frac{-j\pi(1-2k)}{2}}}{j\pi(1-2k)} + \frac{-e^{\frac{-j\pi(1+2k)}{2}} + e^{\frac{j\pi(1+2k)}{2}}}{j\pi(1+2k)} \right)$$

$$= \frac{1}{2j}\left( \frac{e^{\frac{j\pi(1-2k)}{2}} - e^{\frac{-j\pi(1-2k)}{2}}}{\pi(1-2k)} \right) + \frac{1}{2j}\left( \frac{e^{\frac{j\pi(1+2k)}{2}} - e^{-\frac{j\pi(1+2k)}{2}}}{\pi(1+2k)} \right)$$

$$= \frac{1}{\pi(1-2k)} \sin\left[\frac{\pi(1-2k)}{2}\right] + \frac{1}{\pi(1+2k)} \sin\left[\frac{\pi(1+2k)}{2}\right].$$

**Bài 3.4:** Tìm $x(t)$ khi biết các hệ số:

(a) $C_k = \left(\frac{1}{2}\right)^{|k|} e^{jk\frac{\pi}{20}}$. $T=2$.

Thay giá trị $C_k$ và $w_0 = \frac{2\pi}{T} = \pi$ vào $x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jkw_0 t}$.

$$\Rightarrow x(t) = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k e^{jk\frac{\pi}{20}} e^{jk\pi t} + \sum_{k=-1}^{-\infty} \left(\frac{1}{2}\right)^{-k} e^{jk\frac{\pi}{20}} e^{jk\pi t}$$

$$= \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k e^{jk\frac{\pi}{20}} e^{jk\pi t} + \sum_{l=1}^{\infty} \left(\frac{1}{2}\right)^l e^{-jl\frac{\pi}{20}} e^{-jl\pi t}$$

$$\Rightarrow x(t) = \frac{1}{1-\frac{1}{2}e^{j(\pi t+\frac{\pi}{20})}} + \frac{1}{1-\frac{1}{2}e^{-j(\pi t+\frac{\pi}{20})}} - 1$$

$$= \frac{2}{2-e^{j(\pi t+\frac{\pi}{20})}} + \frac{2}{2-e^{-j(\pi t+\frac{\pi}{20})}} - 1$$

$$= \frac{3}{5-4\cos(\pi t+\frac{\pi}{20})}.$$

(b) $C_k = -j\,\delta[k-2] + j\,\delta[k+2] + 2\delta[k-3] + 2\delta[k+3]$ $\omega_0 = \pi$.

$\omega_0 = \pi$.

$$x(t) = \sum_{k=-\infty}^{\infty} C_k \cdot e^{j\pi k t}.$$

$$-j\,\delta[k-2] = \begin{cases} 0 & k \neq 2 \\ -j & k = 2. \end{cases}$$

$$j\,\delta[k+2] = \begin{cases} 0 & k \neq -2. \\ j & k = -2. \end{cases}$$

$$2\delta[k-3] = \begin{cases} 0 & k \neq 3 \\ 2. & k = 3. \end{cases}$$

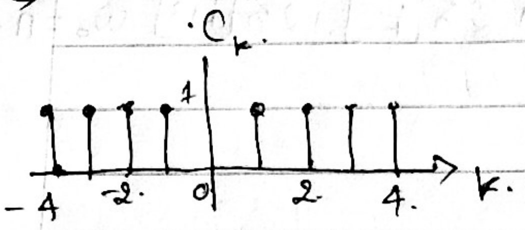
$$2\delta[k+3] = \begin{cases} 0 & k \neq -3 \\ 2. & k = -3. \end{cases}$$

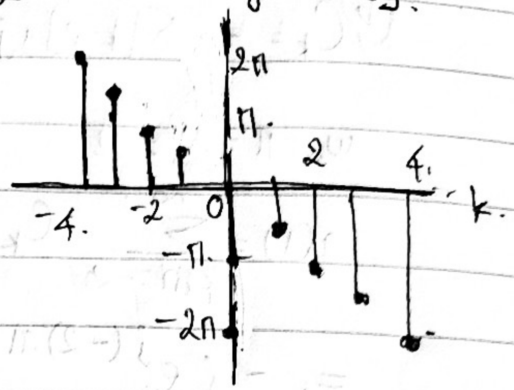
$$\Rightarrow x(t) = -j\,e^{j(2)\pi t} + j\,e^{j(-2)\pi t} + 2 \cdot e^{j(3)\pi t} + 2 \cdot e^{j(-3)\pi t}.$$

$$= -j\left(e^{j2\pi t} - e^{-j2\pi t}\right) + 2\left(e^{j3\pi t} + 2e^{-j3\pi t}\right)$$

$$= -2\sin(2\pi t) + 4\cos(3\pi t).$$

* ⓒ $C_k$: được cho như hình sau, $w_0 = \frac{\pi}{2}$.




$C_k = e^{-j2\pi k} \quad -4 \le k \le 4.$

$\Rightarrow x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk t w_0}$

$= \sum_{k=-4}^{4} e^{-j2\pi k} \cdot e^{jk\frac{\pi}{2}t} = \sum_{-4}^{4} e^{jk(\frac{\pi}{2} - 2\pi t)}$

$= \sum_{k=-4}^{4} e^{jk \cdot 2\pi (\frac{1-4t}{4})}$

Bài 3.5. Tìm biểu diễn Fourier của $x(t)$ cho bởi:

$$x(t) = \begin{cases} 1 & -T_0 \le t \le T_0 \\ 0 & |t| > T_0 \end{cases}$$

Giải:

$$X(w) = F[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-jwt} dt$$

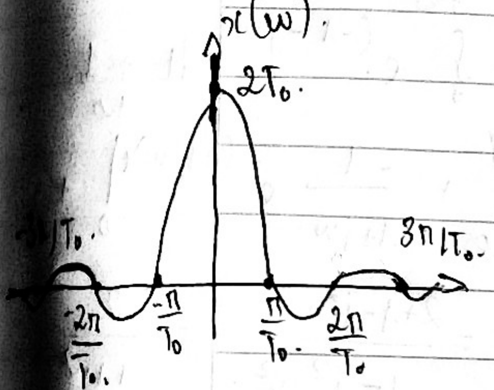
$$= \int_{-T_0}^{T_0} e^{-jwt} dt + \int_{-\infty}^{-T_0} 0\, dt + \int_{T_0}^{+\infty} 0\, e^{-jwt} dt$$

$$= \frac{-1}{jw} e^{-jwt} \Big|_{-T_0}^{T_0} + 0 + 0$$

$$= \frac{-1}{jw} e^{-jwT_0} + \frac{1}{jw} e^{jwT_0}$$

$$= \frac{2}{w} \frac{1}{2j} \left( e^{jwT_0} - e^{-jwT_0} \right)$$

$$= \frac{2}{w} \sin(wT_0).$$




Vậy biểu diễn Fourier của $x(t)$ là

$$X(w) = \frac{2}{w} \sin(wT_0).$$

$w = 0 \Rightarrow \lim_{w \to 0} \frac{2}{w} \sin(wT_0) = 2T_0.$

Bài 3.6. Tìm biểu diễn Fourier của $x(t)$ sau:

a) $x(t) = e^{2t} u(-t)$

$$\Rightarrow x(t) = \begin{cases} 0 & t > 0 \\ e^{2t} & t \le 0 \end{cases}$$

$\Rightarrow$ Biểu diễn Furier là:

$$X(w) = F[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-jwt} dt$$

$$= \int_{-\infty}^{0} e^{2t} e^{-jwt} dt + \int_{0}^{+\infty} 0\, e^{-jwt} dt$$

$$= \int_{-\infty}^{0} e^{-(jw-2)t} dt + 0$$

$$= \frac{-1}{jw-2} e^{-(jw-2)t} \Big|_{-\infty}^{0}$$

$$= \frac{-1}{\ldots}$$

(b) $x(t) = e^{-|t|} = \begin{cases} e^{-t} & t > 0 \\ e^{t} & t < 0 \\ 1 & t = 0 \end{cases}$

$\Rightarrow$ Biểu diễn Furier của $x(t)$ là:

$X(w) = F[x(t)] = \int_{-\infty}^{+\infty} x(t) e^{-jwt} dt$

$= \int_{-\infty}^{0} e^{t} e^{-jwt} dt + \int_{0}^{+\infty} e^{-t} e^{-jwt} dt$

$= \int_{-\infty}^{0} e^{(1-jw)t} dt + \int_{0}^{+\infty} e^{(-1-jw)} dt$

$= \frac{1}{1-jw} e^{(1-jw)t}\Big|_{-\infty}^{0} + \frac{-1}{1+jw} e^{-(1+jw)t}\Big|_{0}^{+\infty}$

$= \frac{1}{1-jw} + \frac{1}{1+jw} = \frac{2}{1-j^2w^2} = \frac{2}{1+w^2}$

(c) $x(t) = e^{-2t} u(t-1)$

$= \begin{cases} e^{-2t} & t \geq 1 \\ 0 & t < 1 \end{cases}$

$\Rightarrow$ Biểu diễn Furier của $x(t)$ là:

$X(w) = F[x(t)] = \int_{-\infty}^{+\infty} x(t) e^{-jwt} dt$

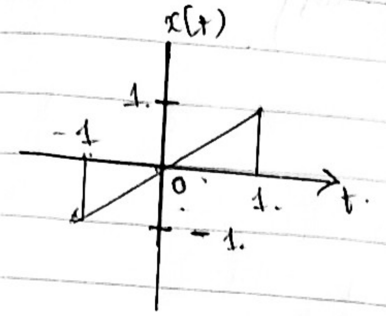
$= \int_{-\infty}^{1} 0\, e^{-jwt} dt + \int_{1}^{+\infty} e^{-2t} e^{-jwt} dt$

$= 0 + \int_{1}^{+\infty} e^{-(jw+2)t} dt$

$= \frac{-1}{jw+2} e^{-(jw+2)t}\Big|_{1}^{+\infty}$

$= 0 + \frac{e^{-(jw+2)}}{jw+2} = \frac{e^{-(jw+2)}}{jw+2}$

d) $x(t)$ cho bởi:




$$X(w) = \int_{-\infty}^{\infty} x(t) e^{-jwt}\, dt$$

$$= \int_{-1}^{0} t e^{-jwt}\, dt + \int_{0}^{1} t e^{-jwt}\, dt$$

$$= I_1 + I_2.$$

$\int t e^{-jwt} = ?$ $\begin{cases} u = t \\ dv = e^{-jwt} \end{cases} \Rightarrow \begin{cases} du = dt \\ v = \dfrac{-e^{-jwt}}{jw} \end{cases}$

$$\Rightarrow \int t e^{-jwt} = t \cdot \frac{-e^{-jwt}}{jw} - \int -\frac{e^{-jwt}}{jw}\, dt$$

$$= \frac{-te^{-jwt}}{jw} + \frac{1}{jw}\int e^{-jwt}\, dt = \frac{-te^{-jwt}}{jw} - \frac{1}{(jw)^2} e^{-jwt}.$$

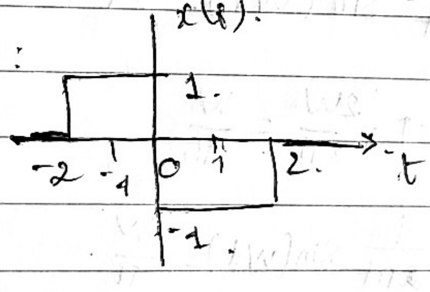
$$\Rightarrow I_1 = 0 - \frac{1}{(jw)^2} - \frac{e^{jw}}{jw} + \frac{e^{jw}}{(jw)^2} + - \frac{e^{-jw}}{jw} - \frac{e^{-jw}}{(jw)^2} - 0 + \frac{1}{(jw)^2}$$

$$= \frac{-1}{jw}\left(e^{jw} + e^{-jw}\right) + \frac{1}{(jw)^2}\left(e^{jw} - e^{-jw}\right)$$

$$= \frac{j}{w}\left(e^{jw} + e^{-jw}\right) - \frac{1}{w^2}\left(e^{jw} - e^{-jw}\right) = \frac{2j}{w}\cos w - \frac{2j}{w^2}\,\frac{1}{2j}\left(e^{jw} - e^{-jw}\right)$$

$$= \frac{2j}{w}\cos w - \frac{2j}{w^2}\sin w.$$

e) $x(t)$ cho bởi:




$$X(w) = \int_{-\infty}^{\infty} x(t) e^{-jwt}\, dt$$

$$= \int_{-2}^{0} e^{-jwt}\, dt - \int_{0}^{2} e^{-jwt}\, dt$$

$$= \frac{-1}{jw} e^{-jwt}\Big|_{-2}^{0} + \frac{1}{jw} e^{-jwt}\Big|_{0}^{2}$$

$$= \frac{-1}{jw} + \frac{1}{jw} e^{+2jw} + \frac{1}{jw} e^{-2jw} - \frac{1}{jw}$$

$$= \frac{-2}{jw} + \frac{1}{jw} e^{2wj} + \frac{1}{jw} e^{-2wj}$$

$$= \frac{-2}{jw}\left(1 + \frac{1}{2}\cdot e^{2wj} + \frac{1}{2}\cdot e^{-2wj}\right).$$

$$= \frac{2\, j^2}{jw}(1 - \cos 2w)$$

$$= \frac{2j}{w}(1 - \cos 2w).$$

$$= \frac{2j(1-\cos 2w)}{w}.$$

Bài 3.7. Biến đổi Furier ngược:

$$X(w) = \begin{cases} 1 & -W < w < W \\ 0 & |w| > W \end{cases}$$

Giải: $x(t) = \frac{1}{2\pi}\cdot\int_{-W}^{W} e^{jwt}\,dw$.

$$= \frac{1}{2j\pi t}\, e^{jwt}\Big|_{w=-W}^{w=W}$$

$$= \frac{1}{\pi t}\sin(Wt). \quad t \neq 0.$$

khi $t = 0$

có $\lim\limits_{t\to 0} \frac{1}{\pi t}\sin(Wt) = \frac{W}{\pi}$.

$$\Rightarrow x(t) = \frac{1}{\pi t}\sin(Wt). = \frac{W}{\pi}\sin\left(\frac{Wt}{\pi}\right).$$

Bài 3.8. Tìm biến đổi Furier ngc của phổ $X(w) = 2\pi\delta(w)$.

$$X(w) = \begin{cases} 2\pi & w = 0 \\ 0 & w \neq 0 \end{cases}$$

$$\Rightarrow x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(w) e^{jwt} dw$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} 2\pi \cdot \delta(w) e^{jwt} dw = \int_{-\infty}^{\infty} \delta(w) e^{jwt} dw$$

$$= e^{j \cdot 0 \cdot t} = e^{0} = 1.$$

$$\boxed{\text{T/c: } \int_{-\infty}^{+\infty} f(t)\delta(t)dt = f(0)}$$

Bài 3.9. Tìm biến đổi Furier ngc của:

(a) $X(w) = \begin{cases} 2\cos w & |w| < \pi \\ 0 & |w| > \pi \end{cases}$

$$\Rightarrow x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(w) e^{jwt} dt$$

$$= \frac{1}{2\pi}\left[\int_{-\pi}^{\pi} 2\cos w \, e^{jwt} dw + \int_{-\infty}^{\pi} 0 \, e^{jwt} dw + \int_{\pi}^{\infty} 0 e^{jwt} dw\right]$$

$$= \frac{1}{2\pi} \cdot 2 \int_{-\pi}^{\pi} \cos w \, e^{jtw} dw$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} \cos w \, e^{jtw} dw.$$

$$I = \int \cos w \, e^{(jt)w} dw = ?$$

$$\begin{cases} u = \cos w \\ dv = e^{jtw} dw \end{cases} \Rightarrow \begin{cases} du = -\sin w \, dw \\ v = \frac{1}{jt} e^{jtw} \end{cases}$$

$$\Rightarrow I = \cos w \frac{e^{jtw}}{jt} + \int \frac{e^{jtw}}{jt} \sin w \, dw.$$

$$= \cos w \frac{e^{jtw}}{e^{jt}} + I_1 \quad I.$$

$$\begin{cases} u = \sin x \\ dv = \frac{e^{jtw}}{jt} dw \end{cases} \Rightarrow \begin{cases} du = \cos dw \\ v = \frac{1}{(jt)^2} e^{jtw} \end{cases} \Rightarrow I_1 = \frac{\sin w \, e^{jtw}}{(jt)^2} - \int \frac{\cos w \, e^{jtw}}{(jt)^2} dw$$

$$\Rightarrow I = \frac{jt\, e^{jtw} \cdot \cos(w) + \sin(w) e^{jtw}}{1+(jt)^2}$$

$$\Rightarrow x(t) = \frac{1}{\pi}\left[\frac{jt\, e^{jtw} \cos(w) + \sin(w) e^{jtw}}{1+(jt)^2}\Bigg|_{-\pi}^{\pi}\right]$$

$$= \frac{1}{\pi}\left[\frac{-jt\, e^{\pi jt}}{1+(jt)^2} - \frac{-jt\, e^{-\pi jt}}{1+(jt)^2}\right]$$

$$= \frac{1}{\pi}\left[\frac{jt\, e^{-\pi jt} - jt\, e^{\pi jt}}{1+(jt)^2}\right]$$

$$= \frac{1}{\pi}\left[\frac{-jt\left(e^{\pi jt} - e^{-\pi jt}\right)}{(1-t)(1+t)}\right]$$

$$= \frac{1}{\pi}\left[\frac{jt\left(e^{\pi jt} - e^{-\pi jt}\right)}{(t-1)(t+1)}\right]$$

$$= \frac{1}{\pi}\left[\frac{\frac{t-1}{2j}\left(e^{j\pi t} + e^{j\pi} - (e^{-j\pi t} + e^{-j\pi})\right) + \frac{t+1}{2j}\left(e^{j\pi t} + e^{-j\pi} - (e^{-j\pi t} + e^{j\pi})\right)}{(t-1)(t+1)}\right]$$

$$= \frac{1}{\pi}\left[\frac{\frac{1}{2j}\left[e^{j\pi(t+1)} - e^{-j\pi(t+1)}\right]}{t+1} + \frac{\frac{1}{2j}\left[e^{j\pi(t-1)} - e^{-j\pi(t-1)}\right]}{t-1}\right]$$

$$= \frac{1}{\pi}\left[\frac{\sin(\pi(t+1))}{t+1} + \frac{\sin(\pi(t-1))}{t-1}\right]$$

**Bài 3.10.**

Đáp ứng xung có dạng

$$h(t) = \frac{1}{RC} e^{-\frac{t}{RC}} u(t)$$

Tìm đáp ứng tần số, đáp ứng pha và đáp ứng biên độ của HT

Giải: $h(t) = \frac{1}{RC} e^{-\frac{t}{RC}} \cdot u(t)$. $\quad u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$

$$\Rightarrow H(w) = \int_{-\infty}^{+\infty} h(t) e^{-jwt} dt$$

$$= \frac{1}{RC} \int_{-\infty}^{\infty} e^{-\frac{\tau}{RC}} u(\tau) \cdot e^{-jw\tau} d\tau$$

$$= \frac{1}{RC} \int_{0}^{\infty} e^{-\frac{\tau}{RC}} e^{-jw\tau} d\tau$$

$$= \frac{1}{RC} \int_{0}^{\infty} e^{-(jw + \frac{1}{RC})\tau} d\tau$$

$$= \frac{1}{RC} \frac{-1}{jw + \frac{1}{RC}} e^{-(jw + \frac{1}{RC})\tau} \Big|_0^{\infty}$$

$$= \frac{1}{RC} \frac{-1}{jw + \frac{1}{RC}} (0 - 1) = \frac{\frac{1}{RC}}{jw + \frac{1}{RC}} = \frac{1}{RCwj + 1}$$

$$= \frac{-RCwj + 1}{(RCw)^2 + 1}$$

$$\Rightarrow H(w) = \frac{-RCw}{(RCw)^2 + 1} j + \frac{1}{(RCw)^2 + 1}$$

$\Rightarrow$ Đáp ứng biên độ: $|H(jw)| = \frac{\frac{1}{RC}}{\sqrt{w^2 + (\frac{1}{RC})^2}}$

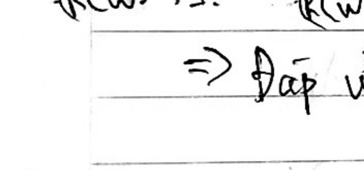
$$= \frac{1}{\sqrt{1 + R^2C^2w^2}}$$

Đáp ứng pha:

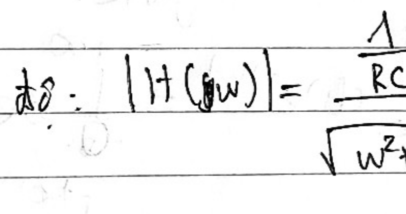
$$\tan \varphi(w) = \frac{Im |H(w)|}{Re |H(w)|}$$

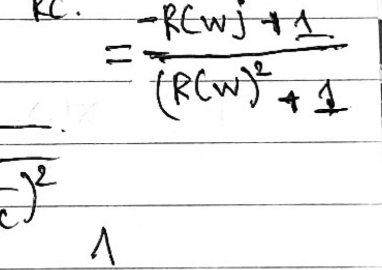
$$= \frac{-RCw}{(RCw)^2 + 1} \cdot \frac{(RCw)^2 + 1}{1} = -RCw$$

$$\Rightarrow \varphi(w) = \arctan(-RCw) = -\arctan(RCw)$$




Mạch RC.



**Bài 3.11.** Xét biến đổi Furier của đáp ứng hệ thống biết:

$$\begin{cases} x(t) = 3e^{-t}u(t). \\ h(t) = 2e^{-2t}u(t) \end{cases}$$

Giải

$$x(t) = \begin{cases} 3e^{-t} & t \geq 0 \\ 0 & t < 0. \end{cases} \Rightarrow X(w) =$$

$$h(t) = \begin{cases} 2e^{-2t} & t \geq 0 \\ 0 & t < 0. \end{cases}$$

$$H(w) = \int_{-\infty}^{\infty} h(t) e^{-jwt} dt.$$

$$= \int_{0}^{+\infty} 2e^{-2t} e^{-jwt} dt.$$

$$= 2\int_{0}^{+\infty} e^{-(jw+2)t} dt = 2 \frac{-1}{jw+2} e^{-(jw+2)t}\Big|_0^{\infty}$$

$$= \frac{2}{jw+2}.$$

$$X(t) = \int_{-\infty}^{\infty} x(t)e^{-jwt} = \int_{-\infty}^{+\infty} 3e^{-t}e^{-jwt}dt$$

$$= \int_{0}^{+\infty} 3e^{-(jw+1)t} dt$$

$$= \frac{-3}{jw+1} e^{-(jw+1)t}\Big|_0^{\infty}$$

$$= \frac{3}{jw+1}.$$

$$\Rightarrow Y(w) = H(w)X(w)$$

$$= \left(\frac{2}{jw+2}\right)\left(\frac{3}{jw+1}\right).$$

Bài 3.12:

Đáp ứng của hệ thống với tín hiệu đầu vào $x(t) = e^{-2t}u(t)$ là $y(t) = e^{-t}u(t)$.

Tìm đáp ứng tần số và đáp ứng xung của hệ thống:

Giải:

$$x(t) = e^{-2t}u(t) = \begin{cases} e^{-2t} & t \geq 0 \\ 0 & t < 0 \end{cases} \Rightarrow X(\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

$$= \int_0^{+\infty} e^{-2t} e^{-j\omega t} dt = \frac{1}{j\omega + 2}$$

$$y(t) = e^{-t}u(t) = \begin{cases} e^{-t} & t \geq 0 \\ 0 & t < 0 \end{cases} \Rightarrow Y(\omega) = \int_{-\infty}^{+\infty} y(t) e^{-j\omega t} dt$$

$$= \int_0^{+\infty} e^{-(j\omega + 1)t} dt = \frac{1}{j\omega + 1}$$

Tìm đáp ứng tần số

$$X(\omega) = \frac{1}{j\omega + 2}$$

$$Y(\omega) = \frac{1}{j\omega + 1}$$

$$\Rightarrow H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{j\omega + 2}{j\omega + 1} = 1 + \frac{1}{1 + j\omega}$$

Tìm đ/ứng xung: $h(t) = F^{-1}[H(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} H(\omega) e^{j\omega t} d\omega$

Sai

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{j\omega + 2}{j\omega + 1} e^{j\omega t} d\omega$$

$$\Rightarrow h(t) = \delta(t) + y(t) = \delta(t) + e^{-t}u(t)$$

* Bài 3.13:

Cho hệ có đáp ứng tần số $H(\omega) = u(\omega + \frac{\pi}{4}) - u(\omega - \frac{\pi}{4})$.

a) Tìm đáp ứng xung.

b) Cho $x(t) = \sin(\frac{t}{2} + \frac{1}{3}) + 2\cos(t - \frac{1}{2}) + 3\sin 2t$.

Tìm đáp ứng của hệ thống:

Giải

$$H(\omega) = u(\omega + \frac{\pi}{4}) - u(\omega - \frac{\pi}{4})$$

$$u\left(w+\frac{\pi}{4}\right)=\begin{cases}0. & w<-\frac{\pi}{4}.\\ 1 & w\geq -\frac{\pi}{4}).\end{cases}$$

$$-u\left(w-\frac{\pi}{4}\right)=\begin{cases}0 & w<\frac{\pi}{4}.\\ -1 & w\geq \frac{\pi}{4}.\end{cases}$$

$$\Rightarrow H(w)=\begin{cases}1. & -\frac{\pi}{4}\leq w<\frac{\pi}{4}.\\ 0 & w<-\frac{\pi}{4}.\\ -1 & w\geq \frac{\pi}{4}.\end{cases}$$

~~b,~~

b, $x(t)=\sin\left(\frac{t}{2}+\frac{1}{3}\right)+2\cos\left(t-\frac{1}{2}\right)+3\sin 2t$

$$=\frac{1}{2j}\left(e^{j\left(\frac{t}{2}+\frac{1}{3}\right)}-e^{-j\left(\frac{t}{2}+\frac{1}{3}\right)}\right)+e^{j\left(t-\frac{1}{2}\right)}+e^{-j\left(t-\frac{1}{2}\right)}$$

$$+\frac{3}{2j}\left(e^{2t}-e^{-2t}\right).$$

Bài 3.9b.

b) $X(w) = 3\delta(w-4) = \begin{cases} 3 & w = 4 \\ 0 & w \neq 4 \end{cases}$

$\Rightarrow x[t] = F^{-1}[X(w)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(w)e^{jwt}\,dw$

$= \frac{1}{2\pi}\cdot 3e^{j4t} + 0 = \frac{3}{2\pi}e^{j4t}$

c) $X(w) = \pi e^{-|w|} = \begin{cases} \pi e^{-w} & w > 0 \\ \pi e^{w} & w < 0 \\ \pi & w = 0 \end{cases}$

$\Rightarrow x(t) = F^{-1}[X(w)] = \frac{1}{2\pi}\left[\int_{-\infty}^{0} \pi e^{+w} e^{jwt}\,dw + \int_{0}^{+\infty} \pi e^{-w} e^{jwt}\,dw\right)$

$= \frac{1}{2}\left[\int_{-\infty}^{0} e^{(jt+1)w}\,dw + \int_{0}^{+\infty} e^{(jt-1)w}\,dw\right)$

$= \frac{1}{2}\left[\frac{1}{jt+1} e^{(jt+1)w}\Big|_{-\infty}^{0} + \frac{1}{1-jt} e^{-(1-jt)w}\Big|_{0}^{+\infty}\right]$

$= \frac{1}{2}\left(\frac{1}{jt+1} \mp \frac{1}{1-jt}\right)$

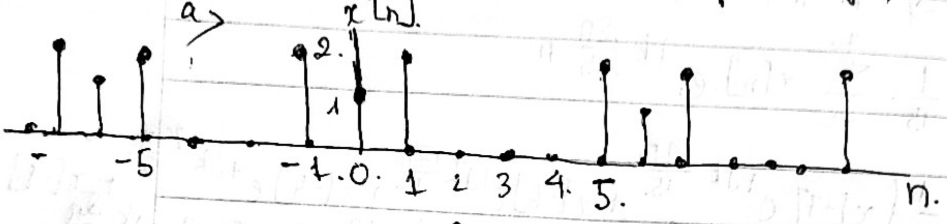
Sai $\begin{cases} = \frac{1}{2}\left(\frac{jt-1+jt+1}{j^2t^2-1}\right) \\ = -\frac{1}{2}\left(\frac{2jt}{t^2+1}\right) = \frac{-1\,jt}{t^2+1} \end{cases}$ Sai.

$= \frac{1}{2}\left(\frac{1-jt-jt-1}{1-(jt)^2}\right) = \frac{1}{2}\cdot\frac{-2jt}{1+t^2} = \frac{-jt}{t^2+1}$

Bài tập chương 3.2.

Bài 1: Xđ phổ pha, phổ biên, phổ công suất các tín hiệu sau.

a)



$N=6 \Rightarrow T = \frac{2\pi}{6} = \frac{\pi}{3}$

$\Rightarrow C_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\Omega_0 n}$

$= \frac{1}{6} \sum_{n=0}^{5} x[n] e^{-jk \cdot \frac{\pi}{3} n}$

$= \frac{1}{6} \{ x[0] \cdot e^{0} + x[1] e^{-j\frac{\pi}{3}k} + x[2] e^{-j\frac{2\pi}{3}k} + x[3] e^{-j\pi k} + x[4] e^{-j\frac{4\pi}{3}k} + x[5] e^{-j\frac{5\pi}{3}k} \}$

$= \frac{1}{6} \{ 1 \cdot 1 + 2e^{-j\frac{\pi}{3}k} + 2e^{-j\frac{5\pi}{3}k} \}$

$= \frac{1}{6} \{ 1 + 2e^{-j\frac{\pi}{3}k} + 2e^{-j\frac{5\pi}{3}k} \} =$

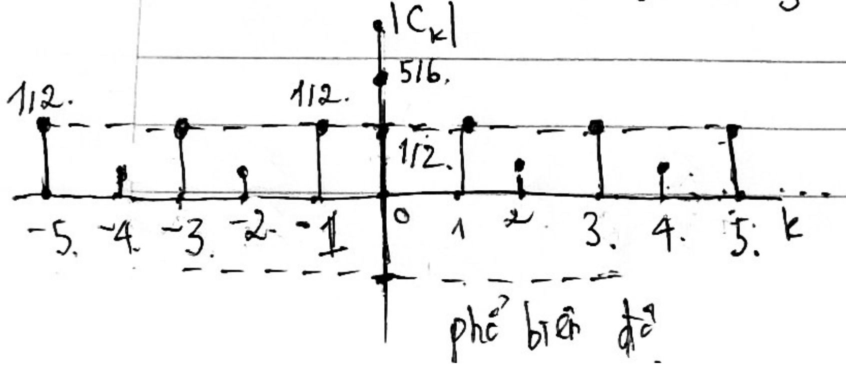
$\boxed{e^{-j\frac{5\pi}{3}k} = e^{j\frac{\pi}{3}k} e^{-j2\pi k} = e^{jk\frac{\pi}{3}}}$

$= \frac{1}{6} + \frac{1}{3} \left( e^{-j\frac{\pi}{3}k} + e^{-j\frac{5\pi}{3}k} \right) = \frac{1}{6} + \frac{1}{3} \left( e^{j\frac{k\pi}{3}} + e^{-j\frac{k\pi}{3}} \right)$

$= \frac{1}{6} + \frac{2}{3} \cdot \frac{1}{2} \left( e^{-j\frac{\pi}{3}k} + e^{-j\frac{5\pi}{3}k} \right) = \frac{1}{6} + \frac{2}{3} \cdot \cos\left(\frac{\pi}{3}k\right)$

$C_0 = \frac{1}{6} + \frac{2}{3} = \frac{5}{6}$ ; $C_1 = \frac{1}{6} + \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{2}$ ; $C_2 = \frac{-1}{6}$.

$C_3 = -\frac{1}{2}$ $C_4 = -\frac{1}{6}$ $C_5 = \frac{1}{2}$.



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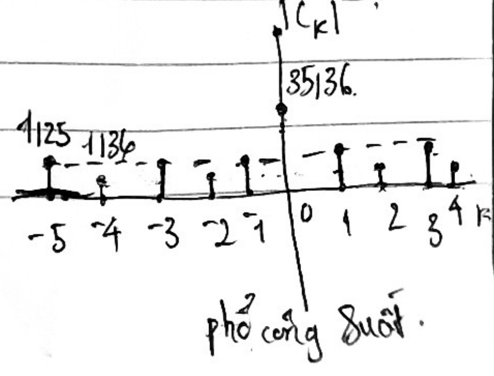


phổ công suất.

b), $N=15 \Rightarrow T=\frac{2\pi}{15}$.

$$C_k = \sum_{-7}^{7} \frac{1}{N}\sum_{N=0}^{N-1} x[n]\, e^{-jk\Omega_0 n}$$

$$= \frac{1}{15}\sum_{-7}^{7} x[n]\, e^{-jk\frac{2\pi}{15}n}$$

$$= \frac{1}{15}\Big\{ x[-7]\, e^{+jk\frac{14\pi}{15}} + x[-6]e^{+jk\frac{12\pi}{15}} + x[-5]e^{+jk\frac{10\pi}{15}} + x[-4]e^{+jk\frac{8\pi}{15}}$$
$$+ x[-3]e^{jk\frac{6\pi}{15}n} + x[-2]\, e^{jk\frac{4\pi}{15}} + x[-1]\, e^{jk\frac{2\pi}{15}} + x[0]e^{0}$$
$$+ x[1]\, e^{-jk\frac{2\pi}{15}} + x[2]\, e^{-jk\frac{4\pi}{15}} + x[3]\, e^{-jk\frac{6\pi}{15}} + \dots + x[7]e^{-jk\frac{14\pi}{15}}$$

$$= \frac{1}{15}\Big\{ 0+0+0+0+0+(-2)\, e^{jk\frac{4\pi}{15}} + (-1)e^{jk\frac{2\pi}{15}} + 0 + e^{-jk\frac{2\pi}{15}} + 2e^{-jk\frac{4\pi}{15}}$$
$$+0+0+0+0+0\Big\}$$

$$= \frac{1}{15}\Big\{ +2e^{-jk\frac{4\pi}{15}} - 2e^{jk\frac{4\pi}{15}} + e^{-jk\frac{2\pi}{15}} - e^{jk\frac{2\pi}{15}}\Big\}$$

$$= \frac{-1}{15}\Big\{\Big(2e^{jk\frac{4\pi}{15}} - 2e^{-jk\frac{4\pi}{15}}\Big) + \Big(e^{jk\frac{2\pi}{15}} - e^{-jk\frac{2\pi}{15}}\Big)\Big\}$$

$$= \frac{-1}{15}2j\Big\{\frac{2}{2j}\Big(e^{jk\frac{4\pi}{15}} - e^{-jk\frac{4\pi}{15}}\Big) + \frac{1}{2j}\Big(e^{jk\frac{2\pi}{15}} - e^{-jk\frac{2\pi}{15}}\Big)\Big\}$$

$$= \frac{-2j}{15}\Big\{ 2.\sin\Big(\frac{4k\pi}{15}\Big) + \sin\Big(\frac{k2\pi}{15}\Big)\Big\}$$

$\Rightarrow C_{-7} =$

**Bài 2**: Sd pt khai triển xđ $C_k$ của tín hiệu:

a> $x[n] = 1 + \sin\left(\frac{\pi n}{12} + \frac{3\pi}{18}\right)$.

$\Omega_0 = \frac{\pi}{12} \Rightarrow N = 24$.

$\Rightarrow x[n] = \sum_{k=0}^{N-1} C_k e^{jk\frac{\pi}{12}n} = \sum_{k=0}^{23} C_k e^{k\frac{\pi}{12}n} = \sum_{k=-11}^{12} C_k e^{k\frac{\pi}{12}n}$.

Sd CT Euler.

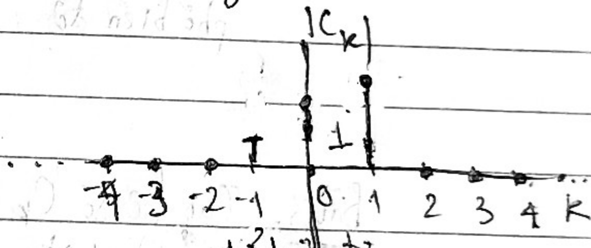
$\Rightarrow x[n] = 1 + \frac{1}{2j}\left(e^{j\left(\frac{\pi n}{12} + \frac{3\pi}{18}\right)} - e^{-j\left(\frac{\pi n}{12} + \frac{3\pi}{18}\right)}\right)$.

$= 1 + \frac{1}{2j} e^{j\frac{\pi n}{12}} \cdot e^{\frac{3\pi}{18}} - \frac{1}{2j} e^{-j\frac{\pi n}{12}} \cdot e^{-\frac{3\pi}{18}}$.

$= 1 + \frac{1}{2j} e^{\frac{3\pi}{18}} \cdot e^{j\frac{\pi}{12}n\cdot(1)} - \frac{1}{2j} e^{-\frac{3\pi}{18}} \cdot e^{j\frac{\pi}{12}\cdot n(-1)}$

$\Rightarrow C_k = \begin{cases} \frac{e^{3\pi/18}}{2j} & k = 1. \\ \frac{-e^{-3\pi/18}}{2j} & k = -1. \\ 1 & k = 0. \\ 0. & k \in [-11; 12] \end{cases}$

$\Rightarrow |C_k| = \begin{cases} \frac{e^{3\pi/18}}{2} & k = 1. \\ \frac{e^{-3\pi/18}}{2} & k = -1. \\ 1 & k = 0 \\ 0 & k \in [-11, 12]. \end{cases}$



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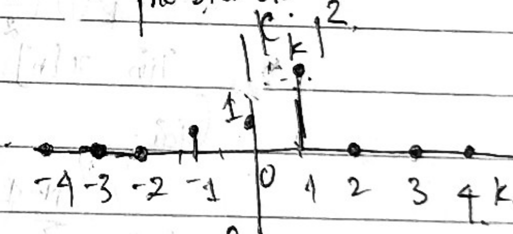


phổ công suất

b> $x[n] = \cos\left(\frac{n\pi}{30}\right) + 2\sin\left(\frac{n\pi}{90}\right)$.

$\ast$ $N_1 = 60$ $N_2 = 180 \Rightarrow N = 180. \Rightarrow w_0 = \frac{2\pi}{180} = \frac{\pi}{90}$.

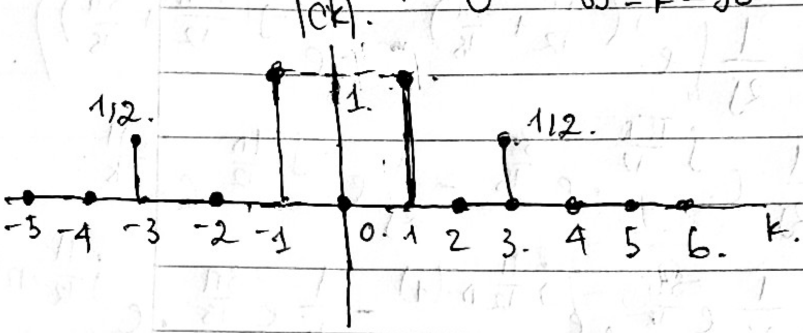
$\Rightarrow x[n] = \sum_{k=0}^{N-1} C_k e^{jk\frac{\pi}{90}\cdot n} = \sum_{k=0}^{179} C_k e^{jk\frac{\pi}{90}n} = \sum_{k=-89}^{90} C_k e^{jk\frac{\pi}{90}n}$

Sd CT Euler.

$$\Rightarrow x[n] = \frac{1}{2}\left(e^{j\frac{\pi n}{30}} + e^{-j\frac{\pi n}{30}}\right) + \frac{1}{j}\left(e^{j\frac{\pi n}{90}} - e^{-j\frac{\pi n}{90}}\right).$$

$$= \frac{1}{2} e^{j\frac{\pi}{90}\cdot n\cdot 3} + \frac{1}{2} e^{j\frac{\pi}{90}\cdot n\cdot(-3)} + \frac{1}{j} e^{j\frac{\pi n}{90}} - \frac{1}{j} e^{j\frac{\pi n}{90}(-1)}$$

$$\Rightarrow C_k = \begin{cases} \frac{1}{j} & k = 1 \\ -\frac{1}{j} & k = -1. \\ \frac{1}{2} & k = \pm 3. \\ 0 & -89 \le k \le 90. \end{cases}$$

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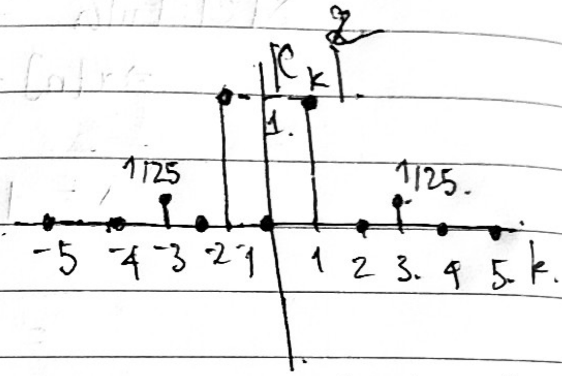



phổ ~~biên độ~~ công suất

**Bài 3.** Cho hệ số $C_k$ của TH cho bởi

$$C_k = \left(\frac{1}{2}\right)^k \quad 0 \le k \le 9.$$

Tìm $x[n]$ với $N = 10$ $\Rightarrow \Omega_0 = \frac{\pi}{5}$.

Giải:

$$x[n] = \sum_{k=0}^{N-1} C_k \cdot e^{jk\frac{\pi}{5}n} = \sum_{k=0}^{9} \left(\frac{1}{2}\right)^k e^{jk\frac{\pi}{5}n}$$

$$k=0 \Rightarrow u_1 = \left(\frac{1}{2}\right)^0 e^0 = 1. \quad k=1 \Rightarrow u = \frac{1}{2} \cdot e^{j\frac{\pi}{5}n}$$

$$\Rightarrow x[n] = \frac{1 - \left(\frac{1}{2}\right)^{10}}{1 - \frac{1}{2} e^{j\left(\frac{\pi}{5}\right)n}}$$

Bài 4: Xđ $x[n]$ ứng với các hệ số chuỗi Fourier.

(CT Euler)

$$C_k = X[k] = \cos\left(\frac{4k\pi}{11}\right) + 2j\sin\left(\frac{6k\pi}{11}\right) = \frac{1}{2}\left(e^{j\frac{4k\pi}{11}} + e^{-j\frac{4k\pi}{11}}\right) + \left(e^{j\frac{6k\pi}{11}} - e^{-j\frac{6k\pi}{11}}\right)$$

$$\Rightarrow C_k = \frac{1}{2}e^{j\frac{4k\pi}{11}} + \frac{1}{2}e^{-j\frac{4k\pi}{11}} + e^{\frac{j6k\pi}{11}} - e^{\frac{-j6k\pi}{11}}$$

$N_1 = \frac{11}{2}$; $N_2 = \frac{11}{3}$ $\Rightarrow N = BCNN(N_1, N_2) = 11$.

$$\Rightarrow \Omega_0 = \frac{2\pi}{N} = \frac{2\pi}{11}$$

Chuỗi Fourier

$$\Rightarrow C_k = \frac{1}{N}\sum_{n=0}^{N-1} x(n).e^{-jk\Omega_0 n} = \frac{1}{11}\sum_{n=0}^{10} x(n).e^{-jk\frac{2\pi}{11}n}$$

$$\Rightarrow C_k = \sum_{n=-5}^{5} x(n)\,e^{-jk.\frac{2\pi}{11}n}$$

$$\Rightarrow C_k = x(-5).e^{-j\frac{2\pi k(-5)}{11}} + x(-4)e^{-j\frac{2\pi k}{11}.(-4)} + x(-3)e^{-j\frac{2\pi k}{11}(-3)}$$
$$+ x(-2).e^{-j\frac{2\pi k}{11}(-2)} + x(-1)e^{-j\frac{2\pi k}{11}.(-1)} + x(1)e^{-j\frac{2\pi k}{11}(1)}$$
$$+ x(2)e^{-j\frac{2\pi k}{11}.2} + x(3)e^{-j\frac{2\pi k}{11}.(3)} + x(4)e^{-j\frac{2\pi k}{11}.(4)}$$
$$+ x(5)e^{-j\frac{2\pi k}{11}.(5)}$$

$$\Rightarrow \begin{cases} x(-2) = \frac{1}{2};\ x(2) = \frac{1}{2};\ x(3) = 1\ \ x(-3) = -1 \\ x(-5) = x(-4) = x(4) = x(5) = x(0) = x(-1) = x(1) = 0 \end{cases}$$

$$\Rightarrow x[n] = \begin{cases} \frac{1}{2} & n = \pm 2 \\ 1 & n = 3 \\ -1 & n = -3 \\ 0, & n \neq\ \ -5 \le n \le 5 \end{cases}$$

**Bài 5: Tìm biến đổi Fourier** $x[n] = 2.3^n u[-n]$

$$x[n] = 2.3^n u[-n] = \begin{cases} 2.3^n & n \le 0 \\ 0 & n > 0 \end{cases}$$

$$\Rightarrow X(\Omega) = F[x(n)] = \sum_{n=-\infty}^{\infty} x(n) e^{-jn\Omega}$$

$$= \sum_{-\infty}^{0} 2.3^n e^{-jn\Omega} + \sum_{0}^{+\infty} 0\, e^{-jn\Omega}$$

$$= 2\sum_{-\infty}^{0} 3^n e^{-jn\Omega} = 2\sum_{-\infty}^{0} \left(3e^{-j\Omega}\right)^n$$

$$= \frac{2.1}{1 - \frac{e^{j\Omega}}{3}}$$

**Bài 6: Tìm biến đổi Fourier của:**

a) $x[n] = \begin{cases} 2^n & 0 \le n \le 9 \\ 0 & n \text{ khác} \end{cases}$

$$X(\Omega) = F[x(n)] = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$$

$$= \sum_{n=0}^{9} 2^n e^{-j\Omega . n} + \sum_{9}^{+\infty} 0 + \sum_{-\infty}^{0} 0 = \sum_{n=0}^{9} (2e^{-j\Omega}$$

$$q = \frac{2^{n+1} . e^{-j\Omega(n+1)}}{2^n e^{-j\Omega n}} = \frac{2.e}{1} = 2e^{-j\Omega}$$

$$\Rightarrow X(\Omega) = \frac{1.(1 - 2^{10} e^{-10j\Omega})}{1 - 2e^{-j\Omega}}$$

b) $x[n] = a^{|n|} \quad a<1.$

$$= \begin{cases} a^{n} & a<1 \quad n>0 \\ a^{-n} & a<1 \quad n<0. \\ 1 & a<1 \quad n=0. \end{cases}$$

$$X(\Omega) = \sum_{n=0}^{\infty} (a^n e^{-j\Omega n}) + \sum_{n=-1}^{-\infty} (a^{-n} e^{-jn\Omega})$$

$$= \sum_{n=0}^{\infty} (a e^{-j\Omega})^n + \sum_{n=-1}^{\infty} (a e^{j\Omega})^{-n}$$

$$= \frac{1}{1-ae^{-j\Omega}} + \frac{ae^{j\Omega}}{1-ae^{+j\Omega}}$$

$$= \frac{1-ae^{+j\Omega} + ae^{j\Omega} - ae^{-j\Omega} \cdot ae^{j\Omega}}{(1-ae^{-j\Omega})(1-ae^{j\Omega})}$$

$$= \frac{1-a^2}{1-(ae^{-j\Omega}+ae^{j\Omega})+a^2}$$

$$= \frac{1-a^2}{1+a^2-2\cdot\frac{1}{2}(ae^{j\Omega}+ae^{-j\Omega})} = \frac{1-a^2}{1+a^2-2a\cos\Omega}$$

c) $x[n] = \delta[6-2n] + \delta[6+2n]$

$$\delta[6-2n] = \begin{cases} 1 & n=3. \\ 0 & n\neq 3. \end{cases}$$

$$\delta[6+2n] = \begin{cases} 1 & n=-3. \\ 0 & n\neq -3. \end{cases}$$

$$\Rightarrow x[n] = \begin{cases} 1 & n=3 \\ 0 & n \text{ khác.} \\ 1 & n=-3. \end{cases}$$

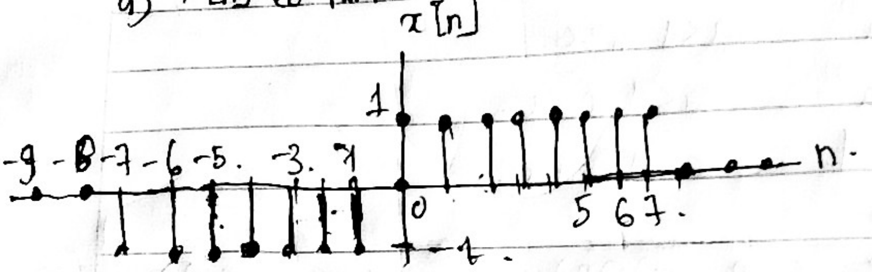
$$\Rightarrow X(\Omega) = F[x(n)] = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$$

$$\left(= \sum_{n=-3}^{3} 2e^{-j\Omega n}\right) \text{Sai}$$

$$= 2\cdot\frac{1}{2}(e^{+3j\Omega} + e^{-3j\Omega}) \cancel{+e^{j\Omega} + e^{-3j\Omega} + e^{-2j\Omega} + e^{j\Omega})}$$

$$= 2\cos 3\Omega.$$

d) $x[n]$ có hình




$$\Rightarrow \begin{cases} x[n] = 1 & -7 \le n \le 7 \;/\{0\} \\ x[n] = 0 & n \text{ khac} \end{cases}$$

$$\Rightarrow X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\Omega n}$$

$$= \sum_{n=-7}^{7} x[n]\, e^{-j\Omega n} + \sum_{-\infty}^{-7} 0 + \sum_{7}^{+\infty} 0$$

$$= -\left(e^{+7\Omega j} + e^{6\Omega j} + e^{5\Omega j} + e^{4\Omega j} + e^{3\Omega j} + e^{2\Omega j} + e^{j\Omega}\right) + \left(e^{-7\Omega j} + e^{-6\Omega j} + e^{-5\Omega j} + e^{-4\Omega j} + e^{-3\Omega j} + e^{-2\Omega j} + e^{-\Omega j}\right) + \ldots$$

$$= -\left(e^{7\Omega j} - e^{-7\Omega j}\right) + \ldots$$

$$= -2j \, \frac{1}{2j} \left(e^{7\Omega j} - e^{-7j\Omega}\right) + \ldots$$

$$= -2j\left(\sin 7\Omega + \sin 6\Omega + \sin 5\Omega + \sin 4\Omega + \sin 3\Omega + \sin 2\Omega + \sin \ldots\right.$$

$$= -2j \sin 7\Omega \, \frac{\sin 4\Omega}{\sin \frac{\Omega}{2}}$$

Bài 7: Tìm biến đổi Fourier ngược của TH:

a) $X(\Omega) = 2\cos(2\Omega) = e^{j2\Omega} + e^{-j2\Omega} = \sum_{n=-\infty}^{\infty} x(n) e^{+j\Omega n}$.

$$x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} 2\cos 2\Omega\, e^{j\Omega n}\, d\Omega.$$

Sai

Sai

$$= \frac{1}{2\pi}\left[\int_{-\pi}^{\pi} e^{j2\Omega} e^{jn\Omega}\, d\Omega + \int_{-\pi}^{\pi} e^{-j2\Omega} e^{jn\Omega}\, d\Omega\right]$$

$$= \frac{1}{2\pi}\left[\int_{-\pi}^{\pi} e^{j(n+2)\Omega}\, d\Omega + \int_{-\pi}^{\pi} e^{j(n-2)\Omega}\, d\Omega\right]$$

$$= \frac{1}{2\pi}\cdot\frac{1}{j(n+2)} e^{j(n+2)}$$

$$\Rightarrow x[n] = \begin{cases} 1. & n = \pm 2. \\ 0 & n \text{ khác.} \end{cases}$$

b) $X(\Omega) = \begin{cases} e^{-j4\Omega} & \frac{\pi}{2} < |\Omega| \le \pi. \\ 0 & \text{khác } -\pi < \Omega < \pi. \end{cases}$

$$x[n] = \frac{1}{2\pi}\int_{\pi/2}^{\pi} e^{-j4\Omega} e^{jn\Omega}\, d\Omega + \frac{1}{2\pi}\int_{-\pi}^{-\pi/2} e^{-j\Omega(4-n)}\, d\Omega.$$

$$= \frac{1}{2\pi}\int_{\pi/2}^{\pi} e^{j\Omega(n-4)}\, d\Omega + \frac{1}{2\pi}\int_{-\pi}^{-\pi/2} e^{-j\Omega(4-n)}\, d\Omega$$

$$= \frac{1}{2\pi}\cdot\frac{1}{j(n-4)} e^{j\Omega(n-4)}\Big|_{\pi/2}^{\pi} + \frac{1}{2\pi}\cdot\frac{1}{j(n-4)} e^{j\Omega(n-4)}\Big|_{-\pi}^{-\pi/2}$$

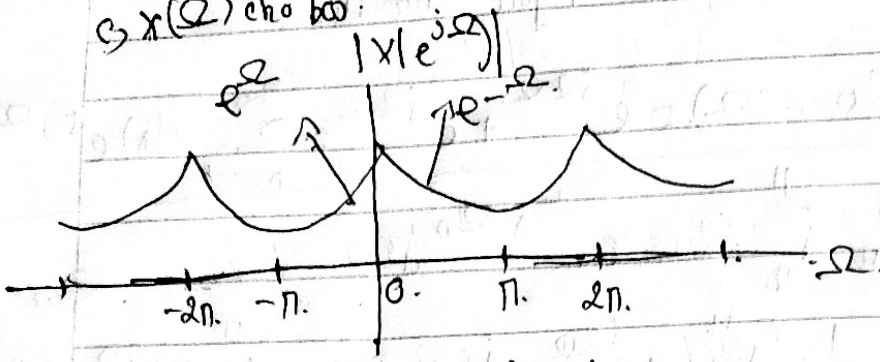
$$= \frac{1}{2\pi}\left(\frac{1}{j(n-4)} e^{j\pi(n-4)} - \frac{1}{j(n-4)} e^{j\frac{\pi}{2}(n-4)} + \right.$$

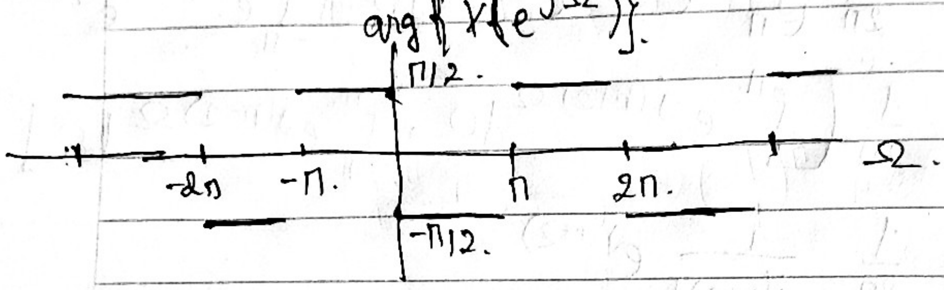
$$\left. + \frac{1}{j(n-4)} e^{-j\frac{\pi}{2}(n-4)} - \frac{1}{j(n-4)} e^{-j\pi(n-4)}\right).$$

$$= \frac{1}{2\pi}\cdot\frac{1}{j(n-4)}\left[\left(e^{j\pi(n-4)} - e^{-j\pi(n-4)}\right) - \left(e^{j\frac{\pi}{2}(n-4)} - e^{-j\frac{\pi}{2}(n-4)}\right)\right]$$

$$= \frac{1}{(n-4)\pi}\left[\sin(\pi(n-4)) - \sin\left(\frac{\pi}{2}(n-4)\right)\right].$$

c) $X(\Omega)$ cho bởi:




$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(W)\, e^{j\Omega n}\, d\Omega$$

$$= \frac{1}{2\pi}\int_{-\pi}^{0} e^{\Omega} e^{j\Omega n}\, d\Omega + \frac{1}{2\pi}\int_{0}^{\pi} e^{-\Omega} e^{j\Omega n}\, d\Omega$$

$$= \frac{1}{2\pi}\int_{-\pi}^{0} e^{\Omega(1+jn)}\, d\Omega + \frac{1}{2\pi}\int_{0}^{\pi} e^{\Omega(jn-1)}\, d\Omega$$

$$= \frac{1}{2\pi}\left( \frac{1}{1+jn} e^{\Omega(1+jn)}\Big|_{-\pi}^{0} + \frac{1}{jn-1} e^{\Omega(jn-1)}\Big|_{0}^{\pi} \right)$$

$$= \frac{1}{2\pi}\left( \frac{1}{1+jn} - \frac{1}{1+jn} e^{-\pi(1+jn)} + \frac{1}{jn-1} e^{\pi(jn-1)} - \frac{1}{jn-1} \right)$$

$$= \frac{+1}{2\pi}\left( \frac{\cancel{1+jn}\; jn-1 - (jn-1)\, e^{-\pi(jn+1)} + (jn+1)\, e^{\pi(jn-1)} + jn\ldots}{j^2 n^2 - 1} \right)$$

$$= \frac{-(2jn - 2}{2\pi(n^2+1)}$$

$$= \frac{1 + e^{-\pi}(-1)^n}{\pi(n^2+1)}$$

Bài 9: Tìm đầu vào x[n] biết đáp ứng và đáp ứng xung HT là:

$$h[n] = \left(\frac{1}{2}\right)^n u[n] = \begin{cases} \left(\frac{1}{2}\right)^n & n \geq 0 \\ 0 & n < 0. \end{cases}$$

$$y[n] = 4\cdot\left(\frac{1}{2}\right)^n u[n] - 2\cdot\left(\frac{1}{4}\right)^n u[n].$$

$$4\cdot\left(\frac{1}{2}\right)^n u[n] = \begin{cases} 4\cdot\left(\frac{1}{2}\right)^n & n \geq 0 \\ 0 & n < 0. \end{cases}$$

$$-2\left(\frac{1}{4}\right)^n u[n] = \begin{cases} -2\left(\frac{1}{4}\right)^n & n \geq 0 \\ 0 & n < 0 \end{cases}$$

$$\Rightarrow y[n] = \begin{cases} 4\cdot\left(\frac{1}{2}\right)^n - 2\cdot\left(\frac{1}{4}\right)^n & n \geq 0 \\ 0 & n < 0. \end{cases}$$

$$= \begin{cases} 4\cdot h[n] - 2\left(\frac{1}{4}\right)^n & n \geq 0 \\ 0 & n < 0. \end{cases}$$

$$Y(\Omega) = 4\left(\frac{1}{1-\frac{1}{2}e^{-j\Omega}}\right) - 2\left(\frac{1}{1-\frac{1}{4}e^{-j\Omega}}\right).$$

$$H(\Omega) = \frac{1}{1-\frac{1}{2}e^{-j\Omega}}.$$

$$X[\Omega] = \frac{Y(\Omega)}{H(\Omega)} = \frac{1-\frac{1}{2}e^{-j\Omega}}{\frac{4}{1-\frac{1}{2}e^{-j\Omega}} - \frac{2}{1-\frac{1}{4}e^{-j\Omega}}}$$

$$\left(= 4.\left(\frac{1}{1-\frac{1}{2}e^{-j\Omega}}\right) \text{Sai}\right.$$

$$= 4 - 2\cdot\frac{1-\frac{1}{2}e^{-j\Omega}}{1-\frac{1}{4}e^{-j\Omega}}.$$

$$\Rightarrow x[n] = 4\delta[n] + 4^n u[n] - \frac{1}{2}\cdot\left(\frac{1}{4}\right)^{n-1} u[n-1]$$